

ELEC5622 Project 1: Step-by-Step Guide

Classifying Alzheimer's disease (AD) patients and normal controls (NC) from brain MRI

Deadline: 11:59pm, Sunday 11 October 2026 (Week 9). Worth 15% of the unit. Registration alone can take several hours for all 50 subjects, so start the pipeline early.

1. What the project is

You have 50 T1-weighted brain MRI scans. Data_00 to Data_39 (training set) are labelled AD or NC in *data_labels.csv*. Data_40 to Data_49 (test set) are unlabelled. Your job is to predict their labels.

This happens in two parts:

- 1 **Measure each brain:** compute the grey matter (GM) volume in each of the first 90 regions of the AAL atlas. Each subject becomes a list of 90 numbers (a 90-D feature vector).
- 2 **Classify:** normalise the features, train a Support Vector Machine (SVM) on the 40 labelled subjects, and predict the 10 test subjects.

The pipeline at a glance

Stage	What it does	FSL tools	Main output (per subject)
Skull stripping	Removes skull and non-brain tissue	bet	Data_XX_brain.nii.gz
Tissue segmentation	Separates CSF / grey / white matter; keeps grey	fast, fslmaths	Data_XX_greymatter_mask.nii.gz
Registration	Warpes the MNI template onto the subject, then carries the AAL atlas along	flirt, fnirt, applywarp	AAL_to_Data_XX.nii.gz
Measurement	GM volume inside each AAL region, written to CSV	fslstats	one row in AAL_statistics_volumnn_*.csv
Classification	Feature scaling + SVM, predict test set	Python (scikit-learn)	predictions.csv, spreadsheet

2. What is in this folder

File	Purpose
SkullStripping	Completed script (stage 1)
TissueSegmentation	Completed script (stage 2)
Registration	Completed script (stage 3)
Measurement	Completed script (stage 4)
CreateSeedMask	Provided helper used by Measurement. Do not modify.
svm.py	SVM classification plus the submission spreadsheet
Project1_Guide.pdf	This document

3. Step-by-step instructions

Step 1: Open the Lab 1 virtual machine

The scripts use FSL, which runs only in the Linux VM you installed in Lab 1, not on Windows.

Step 2: Set up the folders

The scripts expect exactly this layout. All paths are hard-coded to /home/elec5622.

```
/home/elec5622/
|-- SkullStripping  TissueSegmentation  Registration  Measurement  CreateSeedMask
|-- Data/
|   |-- train/      <- Data_00.nii ... Data_39.nii   (40 files)
|   |-- test/       <- Data_40.nii ... Data_49.nii   (10 files)
|-- Packages/       <- aal.nii.gz, MNI152_T1_1mm.nii.gz, MNI152_T1_1mm_brain.nii.gz
|-- Output/
|   |-- train/      (empty)
|   |-- test/       (empty)
```

Commands to create it, after copying the scripts and the Original and Packages folders into the VM:

```
cd /home/elec5622
mkdir -p Data/train Data/test Packages Output/train Output/test
cp /path/to/Original/Data_{00..39}.nii Data/train/
cp /path/to/Original/Data_{40..49}.nii Data/test/
cp /path/to/Packages/*.nii.gz Packages/
chmod +x SkullStripping TissueSegmentation Registration Measurement CreateSeedMask
```

Data/train and Data/test must contain **only** the .nii files, because every script loops over `ls` of those folders.

Step 3: Run skull stripping, then check it

```
./SkullStripping
```

Open a few results in fsleyes, e.g. `fsleyes Data/train/Data_00.nii Output/train/Data_00_brain.nii.gz`, and check that only brain remains.

- Too much brain removed: lower `-f` (e.g. 0.5 to 0.3).
- Skull or neck still visible: raise `-f` (e.g. 0.6).
- Re-run only the bad subject by hand with the new value. Record what you changed, because it belongs in your Methodology section.

Step 4: Run tissue segmentation

```
./TissueSegmentation
```

Check a few `Output/train/Data_XX_greymatter_mask.nii.gz` files overlaid on the brain image. The mask should follow the cortex.

Step 5: Run registration (slow)

```
./Registration
```

`fnirt` takes a few minutes per subject, so the whole run can take hours. Leave it running, ideally overnight. Afterwards, overlay `AAL_to_Data_XX.nii.gz` on the subject's brain in fsleyes (colour map "random") to check that the regions line up.

Step 6: Run measurement

```
./Measurement
```

This produces `Output/train/AAL_statistics_volumn_train.csv` (40 rows) and `Output/test/AAL_statistics_volumn_test.csv` (10 rows). Each row is the subject name followed by one GM volume (mm^3) per AAL region.

If you need to re-run Measurement, delete both CSVs first, because the script appends to them.

Step 7: Run the SVM

Put the two CSVs, `data_labels.csv` (from Original) and `svm.py` in one folder, then run:

```
pip install pandas scikit-learn openpyxl
python svm.py
```

It prints the best SVM settings, the leave-one-out accuracy on the training set, and the 10 predictions. It also writes `predictions.csv` and `ROI_GM_volumes_all_subjects.xlsx`, which is the spreadsheet you submit.

4. What was filled into each script

Each script has a train loop and a test loop. Only the train version is shown below. The test version is identical, with `test_output_path` in place of `train_output_path`. `$filename` is the subject name without its extension, e.g. `Data_00`. Below, `${out}` is shorthand for `${train_output_path}${filename}`.

SkullStripping

```
bet $subID ${out}_brain -f 0.5 -R
```

`-f` sets the fractional intensity threshold (smaller values give a larger brain outline). `-R` runs robust centre estimation, which makes the result more reliable on scans that are not centred.

TissueSegmentation

```
fast -t 1 -n 3 -o $base ${base}_brain
fslmaths ${base}_pve_1 -thr 0.5 -bin ${base}_greymatter_mask
rm -f ${base}_pve_* ${base}_seg.nii.gz ${base}_pveseg.nii.gz ${base}_mixeltype.nii.gz
```

FAST models the brain as 3 tissue classes (`-n 3`) on a T1 image (`-t 1`). Class 1 is grey matter. Its partial-volume map is thresholded at 0.5 and binarised to give the GM mask. The output name `_greymatter_mask` must stay as it is, because Measurement looks for it.

Registration

```
flirt -in $floImg -ref ${out}_brain -omat ${out}_affine.mat # affine (12 DOF)
fnirt --in=$floImg --ref=${out}_brain --aff=${out}_affine.mat --cout=${out}_warp # deformable
applywarp --ref=${out}_brain --in=$aalImg --warp=${out}_warp \
    --out=${train_output_path}AAL_to_${filename} --interp=nn # move the atlas
rm -f ${out}_affine.mat ${out}_warp.nii.gz # clean up
```

The MNI brain template (floating image) is registered to each subject (reference image). The affine step gives a rough global alignment, and `fnirt` refines it non-linearly. The same warp is then applied to the AAL atlas, which is defined in MNI space, so the atlas regions end up in subject space. `--interp=nn` (nearest neighbour) is essential: the atlas holds region labels, and any other interpolation would blend neighbouring labels into meaningless values. The output must be named `AAL_to_<filename>`, because Measurement passes that name to `CreateSeedMask`.

Measurement

```
# inside the region loop (i = 1 .. number of regions)
vol=`fslstats $subGMMask -k ${seedMask_path}${i}.nii.gz -V | cut -d ' ' -f2`
echo -n ", "$vol >> ${train_output_path}"AAL_statistics_volumn_train.csv"
# after the loop: end the subject's row
echo "" >> ${train_output_path}"AAL_statistics_volumn_train.csv"
```

CreateSeedMask splits the warped atlas into one binary mask per region. **fslstats -k** restricts the statistics to that region, and **-V** prints "<voxels> <volume in mm³>". cut keeps the volume. The atlas has up to 116 regions, and svm.py keeps the first 90 as the project requires.

5. How the SVM works (for your Methodology section)

- Features: the first 90 AAL GM volumes per subject.
- **Normalisation:** each feature is z-scored (zero mean, unit standard deviation). Raw volumes differ by orders of magnitude between regions, and SVMs are sensitive to scale. The scaler is fitted on training data only (inside a pipeline), so no information leaks from the validation fold or test set.
- **Model selection:** grid search over kernel (linear, RBF) and C (0.01 to 100), scored with leave-one-out cross-validation (LOOCV). LOOCV suits a small dataset of 40 subjects.
- The best model is refitted on all 40 training subjects and used to predict Data_40 to Data_49.

Ideas for extra marks in Results and Discussion: report a confusion matrix and sensitivity/specificity from LOOCV; compare linear and RBF kernels; for the linear SVM, list the regions with the largest weights and check whether they match known AD atrophy (hippocampus, parahippocampal gyrus, amygdala, temporal lobe); discuss limitations (small sample, no correction for head size / total intracranial volume, registration errors).

6. What to submit

- 1 **Spreadsheet:** 90 ROI GM volumes for all 50 subjects (*ROI_GM_volumes_all_subjects.xlsx* from svm.py). Optionally replace ROI_1..ROI_90 with the AAL region names.
- 2 **Group report**, with team contributions stated at the beginning.

Recommended report structure and marking weights

Section	Content	Marks
1. Aims and Background	What AD is, why MRI GM atrophy is a biomarker, why automated classification matters. Cite literature.	20%
2. Contributions	Who did what.	Part of the above
3. Methodology	Each pipeline stage (sections 4 and 5 of this guide), parameters used, and any manual fixes (e.g. bet -f changes).	30% (together with background knowledge and sources)
4. Results	Best SVM parameters, LOOCV accuracy, confusion matrix, prediction table for Data_40 to Data_49, and example figures (skull strip, GM mask, warped atlas).	20%
5. Discussion and Conclusion	Interpret the results against published findings, limitations and improvements. This section carries the most weight.	30%

Prediction table format (the brief says "Data_30", which is a typo; the test set is Data_40 to Data_49):

File Name	Prediction
Data_40.nii	AD / NC
...	...
Data_49.nii	AD / NC

7. Troubleshooting

Problem	Likely cause and fix
"Permission denied" when running a script	Run <code>chmod +x</code> on the scripts.
"No such file" for <code>_brain / _greymatter_mask / AAL_to_</code>	A previous stage failed or was skipped. Run the stages in order and check the Output folders.
<code>/bin/bash^M: bad interpreter</code>	The file picked up Windows line endings. Run <code>sed -i 's/\r\$//'</code> SkullStripping TissueSegmentation Registration Measurement CreateSeedMask
CSV has duplicated rows	Measurement was run twice. Delete both CSVs and re-run.
Measurement says seedMasks already exists	A previous run was interrupted. Delete <code>Output/train/seedMasks</code> (and test) and re-run.
Very low LOOCV accuracy	Check skull stripping and registration visually. Bad bet results distort the GM volumes.